TourSafe

AI-Powered Travel Safety · For Beginners

An easy guide to understanding what TourSafe is, why it exists, and how it works

25×	30 min	7–10%	70%
More deadly than disease for travelers	Avg. emergency response in remote areas	Survival drop per minute of delay	Response time reduction target

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1. What is the Problem?

Imagine you are hiking alone in a forest. You slip and break your leg. You're unconscious. Your phone is 5 metres away. Nobody knows where you are.

This is the exact situation millions of international travelers face. And the current systems cannot help them.

> **📖 Real-World Example**
> A solo tourist goes trekking in a remote valley in Himachal Pradesh. She suffers a severe fall. There is no network. Nobody sees it happen. The nearest hospital is 40 km away. Even if someone found her after 20 minutes, authorities would not know her blood type, allergies, or who to call. By the time help arrives — it could be too late.

Why is it so bad right now?

Problem 1	Problem 2	Problem 3
No Automatic Detection	**Unknown Identity**	**Slow Response**
If you're unconscious, you can't press SOS. Current systems are useless when you need them most.	6,000+ ID types globally. Police cannot read a foreign ID. They don't know your blood type or allergies.	In remote areas, emergency response takes up to 30 minutes. Each lost minute = 7–10% drop in survival.

⏱ The 'Golden Hour' — Why Every Minute Counts

- Minute 0–5 → Accident happens. Nobody knows.
- Minute 5–20 → Someone finds the victim. No ID. No medical info. Cannot call family.
- Minute 20–30 → Police finally arrive. Paperwork begins. Hospital notified separately.
- Minute 30+ → Help arrives. But survival probability has already dropped 20–30%.
- TourSafe Goal: Compress steps 1–3 into under 5 MINUTES automatically.

2. What Does TourSafe Do?

TourSafe is a smartphone app + server system that watches over travelers silently. It detects accidents automatically, identifies the victim instantly, and calls for help — all without the traveler needing to do anything.

Simple Analogy

Think of TourSafe as a silent guardian angel on your phone. It watches your movement 50 times every second. The moment something goes wrong — a crash, a fall, lying unconscious — it wakes up, identifies you, and sends emergency services to your exact location. Automatically. Even if you are deep in a forest with no network signal.

TourSafe's 3-Layer Shield

Layer 1 — The Watchful Guardian (AI Brain)

The AI that never sleeps, watching your movement 24/7

- Your phone's sensors measure your movement 50 times per second
- An AI model learns what 'normal' movement looks like (walking, driving, sitting)
- If you crash or go unconscious — the AI detects it automatically
- No button press needed. It works even if you're unconscious

Layer 2 — The Bridge of Trust (Blockchain ID)

Your digital identity that any doctor or police officer can read in seconds

- You create a secure digital ID with your blood type, allergies, and emergency contacts
- This ID is locked with a secret key — only you (and emergencies) can open it
- First responders scan a QR code on your phone to instantly see your medical info
- No language barriers. No paperwork. Information available in seconds

Layer 3 — The Command Centre (Dashboard & Maps)

The control room for police and emergency services

- Police and hospitals see a live map of all tracked travelers in their area
- Danger zones (forests, mountains, risky roads) are marked as virtual fences
- When AI detects an emergency, an automatic report is sent to police + hospital instantly
- The report includes the traveler's location, identity, and medical info — no paperwork

3. How Does It Work? (Step by Step)

Here is exactly what happens from the moment a traveler installs TourSafe to the moment help arrives after an accident.

Part A — Setup (Before the Trip)

1

Download & Register

The traveler downloads the TourSafe app. They enter their name, blood type, allergies, medications, and emergency contact (e.g. family member's phone number).

▼

2

Digital ID is Created

TourSafe creates a secure digital identity (called a DID) for the traveler. It is stored on a blockchain — a tamper-proof digital record that cannot be faked or changed.

▼

3

QR Code Generated

The app generates a personal QR code. This QR code is shown on the lock screen of the phone. Any first responder who scans it gets instant access to the traveler's medical info.

Part B — During the Trip (Normal Monitoring)

GPS Tracking (1 time/second)

The app records the traveler's location every second and sends it to the server. The server caches it so the dashboard always shows the traveler's latest position.

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Sensor Monitoring (50 times/second)

The phone's motion sensor captures 50 readings per second. Every 3 seconds, the AI analyses a batch of these readings to check if movement is normal.

▼

Offline Mode (No Signal)

If the phone loses network (common in forests/mountains), TourSafe stores all data encrypted on the phone itself. The moment signal returns, it all syncs automatically — zero data loss.

Part C — Emergency Detected

Crash or Unconsciousness Detected

The AI detects either: (a) a sudden massive G-force spike = crash, OR (b) the phone lying perfectly still at 9.81 m/s^2 for a long time in a remote area = unconscious. It triggers an alert.

▼

Identity Unlocked

The blockchain DID is activated. The traveler's medical info (blood type, allergies, contacts) is decrypted and made available to the emergency team.

▼

e-FIR Auto-Generated

An electronic First Information Report (like a police report) is automatically created with: exact GPS coordinates, victim's medical info, time of incident, and type of emergency.

▼

Police + Hospital Notified

The e-FIR is sent instantly to the nearest police station AND nearest hospital (calculated using GPS math). Their dashboard lights up with the alert. Help is dispatched immediately.

Total Time: Under 5 Minutes

From accident → AI detection → identity unlock → e-FIR generation → police + hospital notified. All automated. No human needed to start the chain.

4. Understanding the AI — Made Simple

What is the AI doing exactly?

The AI in TourSafe is called an LSTM Autoencoder. That sounds scary — but here's a simple way to understand it:

Analogy: The 'Normal Day' Diary

Imagine someone who writes in a diary every day describing what a normal day feels like: walking to college, riding a bus, sitting in class, sleeping. One day, they read an entry and it says 'somersaulted 3 times and hit concrete at 80 km/h.' They immediately know something is terribly wrong — because it doesn't match any normal day. The LSTM Autoencoder works exactly the same way, but with phone sensor data instead of diary entries.

Step-by-Step: How the AI Detects an Accident

Step	What Happens	Simple Explanation
Step 1	Sensor reads Ax, Ay, Az (3 axes of movement)	Like measuring how much you shake left-right, front-back, and up-down
Step 2	Convert to $A\_mag = \sqrt{Ax^2+Ay^2+Az^2}$	Combine all 3 into one total shake number — works no matter how the phone is held
Step 3	Group 3 seconds of readings (150 data points)	Bundle 3 seconds of data together as one 'snapshot' to examine
Step 4	AI tries to 'reconstruct' the snapshot	The AI re-draws what it saw. If it can re-draw it easily = normal. If it struggles = abnormal!
Step 5	Measure the Reconstruction Error	High error = the AI couldn't re-draw it = something unusual happened = potential emergency
Step 6	Two consecutive windows both show high error	Confirm it's not a random shake. Two snapshots in a row must look abnormal before alert fires.

Two Types of Emergencies the AI Catches

Crash / Impact

What the sensor sees:
A massive sudden SPIKE in movement numbers followed by either chaos or dead stillness
Real example:
Car crash → phone flies → huge G-force → AI sees it → ALERT!

Unconsciousness

What the sensor sees:
Phone is PERFECTLY still at exactly 9.81 m/s^2 (Earth's gravity) for a long time in a remote area
Real example:
Hiker collapses unconscious → phone still → AI detects stillness → ALERT!

5. Blockchain Identity — Made Simple

This is the part that solves the 'we don't know who you are' problem. Here's how to understand it:

> 🔐 **Analogy: The Safety Deposit Box**
>
> Imagine a safety deposit box at a bank. You put your medical records inside. You lock it with your private key. The bank (blockchain) has a copy of your box's serial number — but NOT the key. Only when a doctor or police officer is authorized during a real emergency do they get a temporary key to open it. They see what they need. The key expires after the emergency. Your records go back to being locked.

The 4-Step Identity Flow

🔑 Step 1	🔒 Step 2	⛓ Step 3	🚑 Step 4
App generates 2 keys: a PUBLIC key (shareable) and a PRIVATE key (stays on your phone only)	Your medical info is ENCRYPTED with the public key and stored on IPFS (a distributed storage network)	The public key + location of your data is registered on the Polygon Blockchain (permanent, tamper-proof)	Emergency detected → responder scans QR → blockchain releases temporary decrypt key → data shown in seconds
Onboarding	**Encryption**	**Registration**	**Emergency Access**

What Data is in the Digital ID?

Information	Why It Matters in an Emergency
Blood Type (e.g. A+, O-)	Doctors need this BEFORE giving blood transfusions. Wrong blood type can kill.
Allergies (e.g. Penicillin)	Giving the wrong medicine can cause fatal reactions in seconds.
Chronic Conditions (e.g. Diabetes)	Changes how treatment is administered — critical for doctors to know.
Emergency Contact (Name + Phone)	Family is notified immediately. No guesswork about who to call.
Insurance Policy Number	Hospital can verify coverage instantly without calling the insurer manually.
Home Country & Language	Helps responders communicate or get a translator immediately.

6. System Architecture — The Big Picture

Here is how all the pieces of TourSafe connect to each other.

TRAVELER — React Native App (iOS & Android)			
IMU Sensor 50 Hz motion data	**GPS** 1 Hz location updates	**DID Wallet** Encrypted medical ID	**Offline Buffer** SQLite queue when offline

⬇ **WebSocket (Real-time data stream)** ⬇

BACKEND — FastAPI Server (Python) + Redis Cache + MongoDB Database		
LSTM AI Engine Analyses sensor data every 1.5 seconds. Flags anomalies.	**Socket.io Stream** Pushes alerts to dashboard in milliseconds via WebSocket.	**e-FIR Microservice** Auto-generates incident report and dispatches it.

⬇ **Polygon Blockchain** ⬇

Identity Smart Contract Stores DID records. Grants emergency decryption. Immutable audit log.	**B2G Authority Dashboard** Police + hospital see live map, receive e-FIR, coordinate response.

7. Technologies Used — Beginner's Glossary

Here are all the technologies TourSafe uses, explained as simply as possible.

📱 **Mobile App**	React Native (bare CLI) + TypeScript — Build one app for iPhone AND Android. TypeScript catches coding mistakes before they happen.
🗄 **Backend**	FastAPI (Python) — The server that receives data from thousands of phones simultaneously without slowing down. Like a very fast post office.
⚡ **Live Updates**	Redis — Ultra-fast memory storage for live GPS positions. Like a whiteboard that updates in real-time.
💾 **Database**	MongoDB — Stores traveler profiles, incident logs, and e-FIR records. Like a giant, flexible filing cabinet.
🔔 **Alerts**	Socket.io (WebSockets) — Pushes emergency alerts to the police dashboard the millisecond they happen. Like a direct phone line that's always open.
🧠 **AI Engine**	TensorFlow + ONNX Runtime — Trains the accident-detection AI (TensorFlow) and runs it fast in production (ONNX). Like training a dog and then letting it work independently.
⛓ **Blockchain**	Polygon PoS + Solidity + Hardhat — Stores identity records permanently. Polygon is cheap (₹0.001 per transaction vs Ethereum's ₹500+).
🗺 **Maps**	Mapbox GL JS — Renders the live authority map with incident heatmaps. Like Google Maps but programmable for emergency operations.
🖥 **Dev Tools**	VS Code + AnthropicAI + GitHub Copilot — AI-assisted coding. Android Studio — Android phone emulator for testing.
📦 **Deployment**	Docker + Docker Compose — Packages all services into containers so the whole system can run identically on any machine or cloud server.

8. Development Plan — 4 Sprints

TourSafe is built in 4 sprints (phases). Each sprint has specific goals:

Sprint 1 · Core Infrastructure & Scaffolding

- Set up Docker, MongoDB, Redis, FastAPI server
- Build the React Native app skeleton with all screens
- Set up the React.js authority dashboard with Mapbox
- Configure Android Studio emulator for testing

Sprint 2 · Client Telemetry & Offline Caching

- Connect phone sensor (50Hz IMU) and GPS (1Hz) to send live data
- Build the SQLite offline buffer with AES-256 encryption
- Implement Turf.js geo-fence boundary detection on the phone
- Test offline → reconnect → data sync flow

Sprint 3 · AI Execution & Authority Communication

- Train the LSTM Autoencoder on normal movement data
- Export to ONNX Runtime for fast production inference
- Wire Socket.io to push alerts from server to dashboard instantly
- Implement SNN-CAD geographic trajectory hazard detection

Sprint 4 · Blockchain DID & Auto e-FIR

- Write and deploy Identity Resolution smart contract on Polygon Amoy
- Build DID onboarding: key generation, vault encryption, IPFS storage
- Build QR-code emergency decryption flow for first responders
- Build the e-FIR microservice: auto-generate and dispatch reports

9. What Changes With TourSafe?

Here is a direct before-and-after comparison:

Situation	WITHOUT TourSafe	WITH TourSafe
Unconscious traveler found	Nobody knows who they are. Paperwork begins. 30+ minutes wasted.	QR scan → instant ID + blood type + emergency contact in under 15 seconds.
Remote area accident	No signal = no help. Traveler is invisible.	Data cached offline. Syncs instantly when signal returns. No data lost.
Emergency response time	20–30 minutes for identification + dispatch	Under 5 minutes from accident to police + hospital being notified
Police report (FIR)	Officer types it manually after arriving on scene	Auto-generated e-FIR sent to police BEFORE they even leave the station
Crossing a dangerous zone	No warning given to traveler	Instant push notification the moment they enter a geo-fenced hazard zone
Cross-border ID problems	6,000 ID types, language barriers, no medical history	Universal DID on blockchain — readable by any authorized responder worldwide

5,000+	**< 5 min**	**70%**	**₹0**
Travelers protected in initial rollout	Total emergency response time	Reduction in response latency	Cost for travelers (B2G model)

The Mission in One Sentence

TourSafe ensures that no traveler — no matter how remote, how unconscious, or how foreign — is ever truly alone in an emergency.

SDG 8	**SDG 11**	**SDG 16**
Decent Work & Economic Growth — Protects tourism economy	Sustainable Cities — Safer, more resilient communities	Peace & Justice — Faster, accountable emergency institutions

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